

GRASSWORKS: Analysis of Bauer et al. (submitted) Functional traits of grasslands: Community weighted mean of specific leaf area (SLA) compared to reference sites

MB

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NOTE: To compare different models, you only have to change the models in the section ‘Load models’

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```
library(here)
library(tidyverse)
```

Packages

```
## Warning: Paket 'ggplot2' wurde unter R Version 4.5.3 erstellt
```

```
## Warning: Paket 'purrr' wurde unter R Version 4.5.3 erstellt
## Warning: Paket 'dplyr' wurde unter R Version 4.5.3 erstellt
library(ggbeeswarm)
library(patchwork)
library(DHARMA)
library(emmeans)
```

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites_refs.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    eco.id = "f",
    region = col_factor(levels = c("north", "centre", "south"), ordered = TRUE),
    site.type = col_factor(
      levels = c("positive", "restored", "negative"), ordered = TRUE
    ),
    obs.year = "f",
    hydrology = col_factor(levels = c("moist", "fresh", "dry"), ordered = TRUE),
    mngm.type = "f",
  )
) %>%
  rename(y = cwm.abu.sla)
```

Load data

Statistics

Data exploration

Means and deviations

```
Rmisc::CI(sites$y, ci = .95)

##      upper      mean      lower
## 239.7403 237.1109 234.4816

median(sites$y)

## [1] 237.07

sd(sites$y)

## [1] 33.66074

quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)

##      5%      95%
## 181.8215 292.1370

sites %>% distinct(id.site) %>% nrow()

## [1] 172

sites %>% count(eco.id)

## # A tibble: 3 x 2
```

```
##   eco.id      n
##   <fct>   <int>
## 1 654      209
## 2 664      212
## 3 686      211
```

```
sites %>% count(site.type)
```

```
## # A tibble: 3 x 2
##   site.type      n
##   <ord>       <int>
## 1 positive     57
## 2 restored    472
## 3 negative    103
```

```
sites %>% count(hydrology)
```

```
## # A tibble: 3 x 2
##   hydrology      n
##   <ord>       <int>
## 1 moist      135
## 2 fresh      327
## 3 dry        170
```

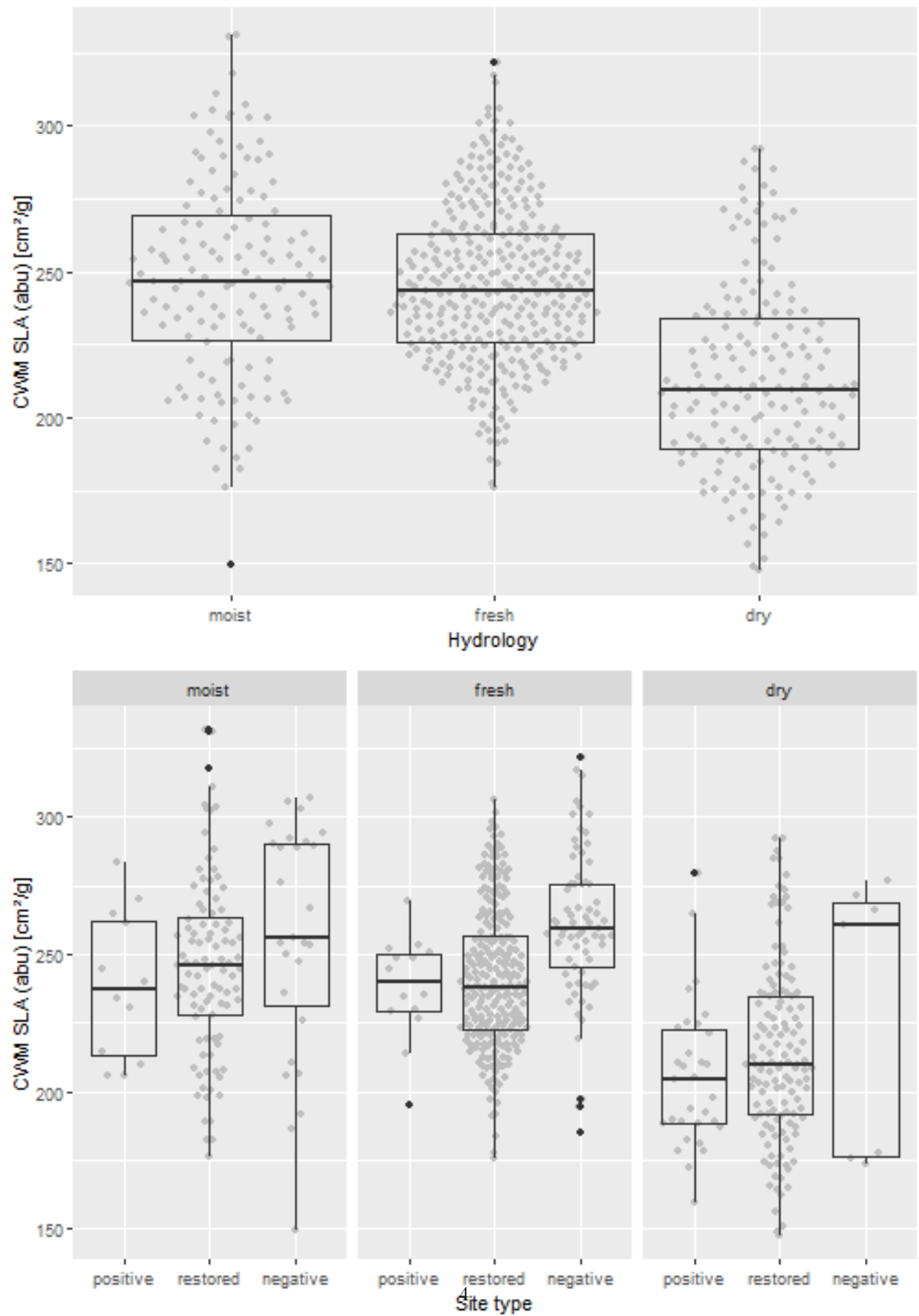
```
sites %>% count(hydrology, eco.id)
```

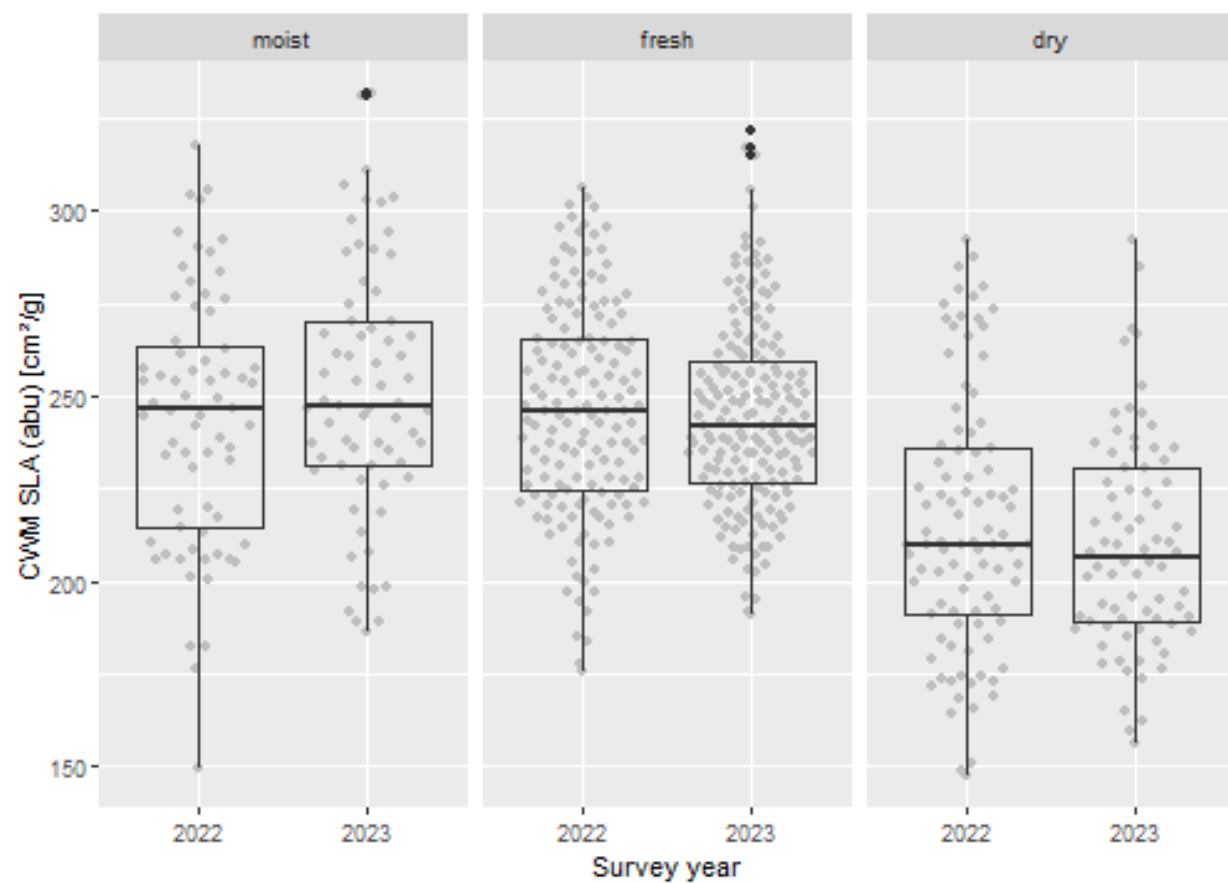
```
## # A tibble: 9 x 3
##   hydrology eco.id      n
##   <ord>     <fct>   <int>
## 1 moist    654      35
## 2 moist    664      50
## 3 moist    686      50
## 4 fresh    654     108
## 5 fresh    664     109
## 6 fresh    686     110
## 7 dry      654      66
## 8 dry      664      53
## 9 dry      686      51
```

```
sites %>% count(hydrology, site.type)
```

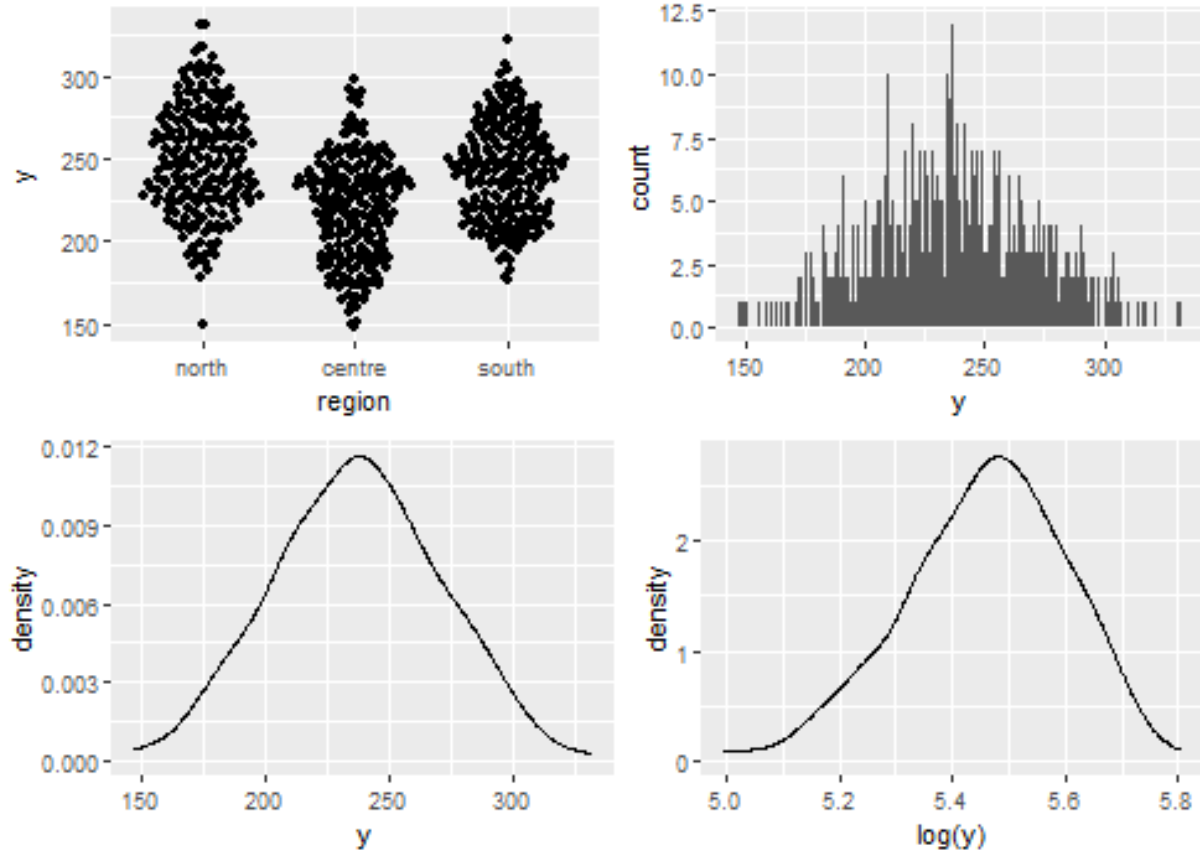
```
## # A tibble: 9 x 3
##   hydrology site.type      n
##   <ord>     <ord>     <int>
## 1 moist    positive    12
## 2 moist    restored    96
## 3 moist    negative    27
## 4 fresh    positive    14
## 5 fresh    restored   244
## 6 fresh    negative    69
## 7 dry      positive    31
## 8 dry      restored   132
## 9 dry      negative     7
```

Graphs of raw data (Step 2, 6, 7)





Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude $r > 0.7$ Dormann et al. 2013 Ecography DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cum.)) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#   ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_sla_refs_1.Rdata"))
load(file = here("outputs", "models", "model_sla_refs_2.Rdata"))
m_1 <- m1
m_2 <- m2

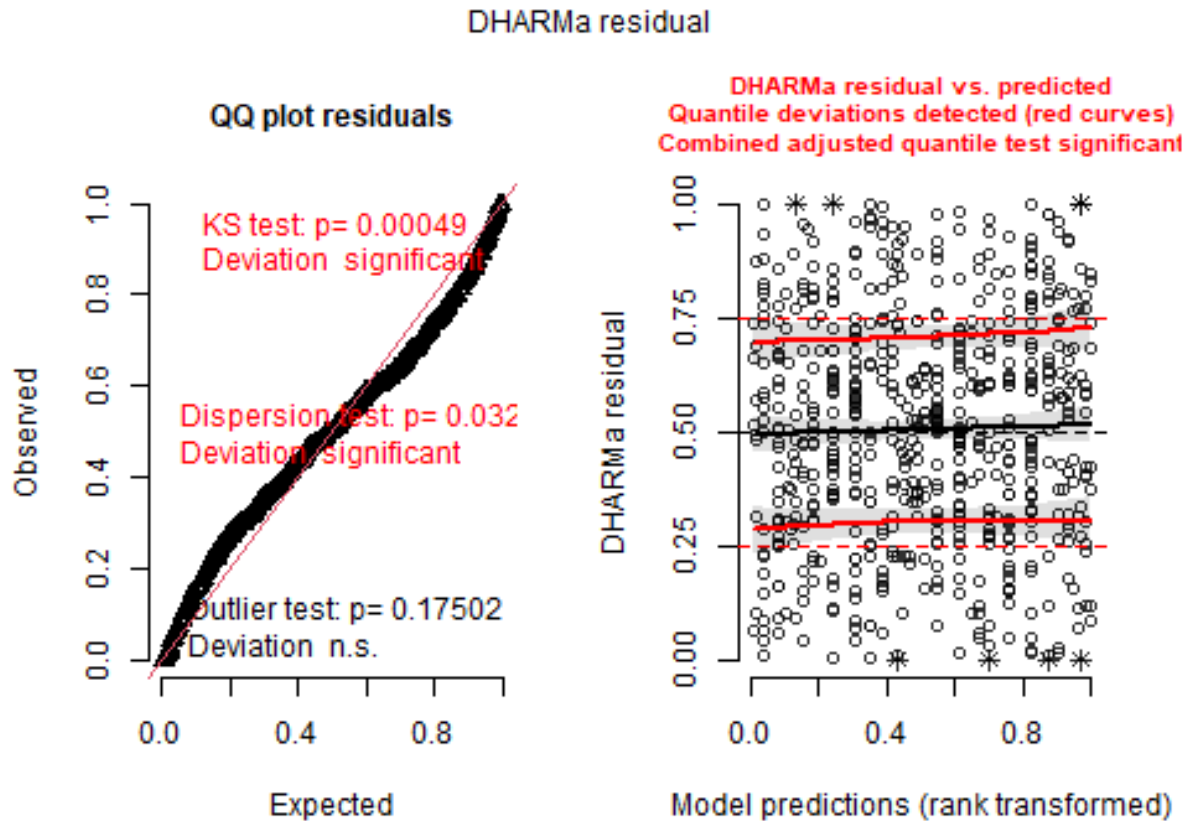
m_1@call
## lmer(formula = sqrt(y) ~ hydrology * site.type + eco.id + obs.year +
##       (1 | id.site), data = sites, REML = FALSE)
```

```
m_2@call
## lmer(formula = y ~ (hydrology + site.type + eco.id + obs.year)^2 +
##       (1 | id.site), data = sites, REML = FALSE)
```

Model check

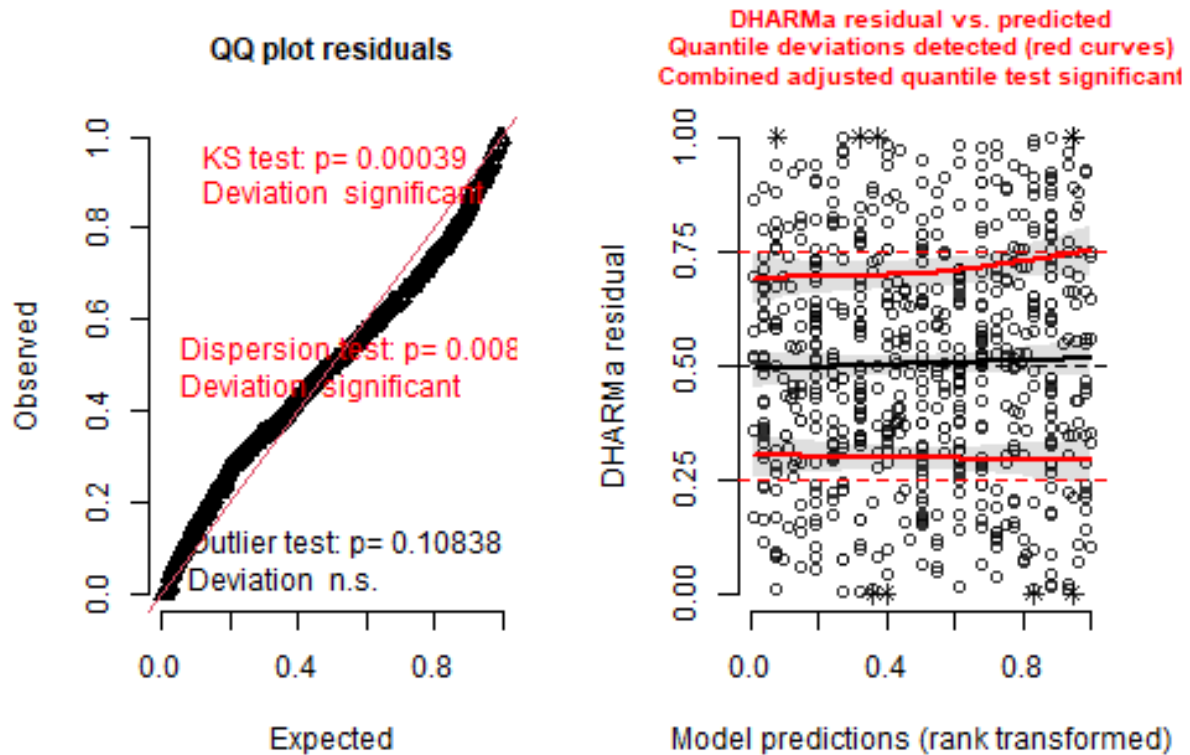
DHARMA

```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```

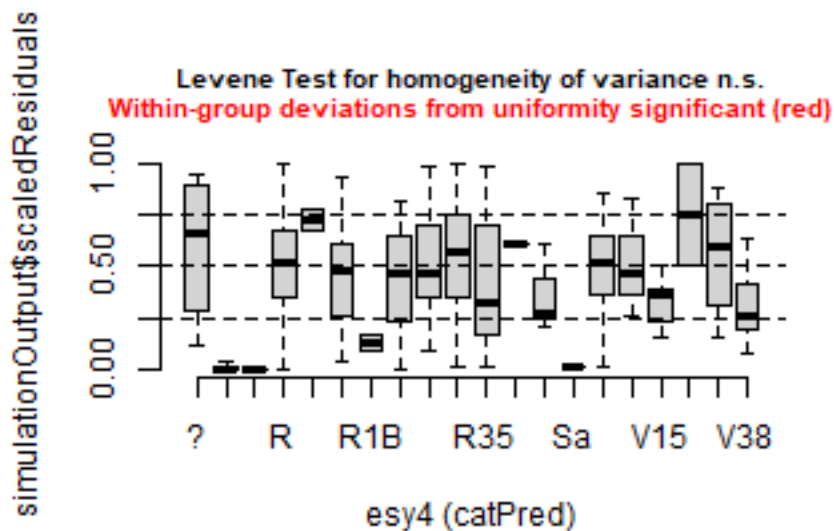


```
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```

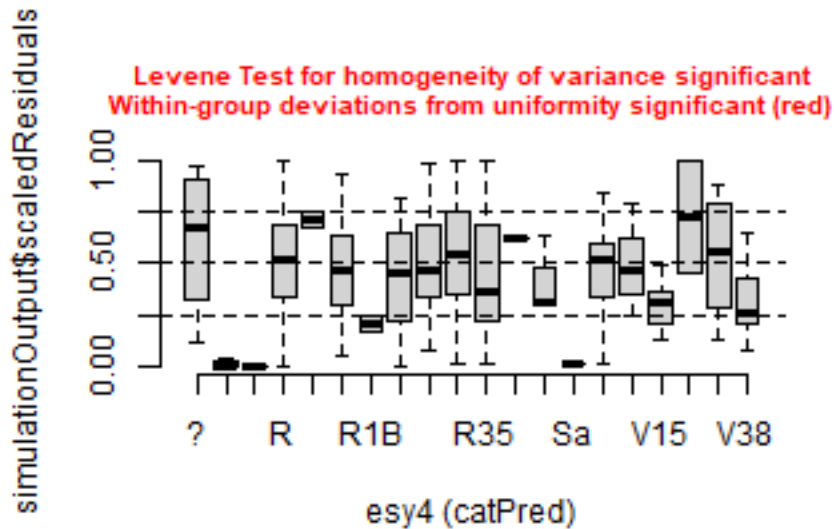
DHARMa residual



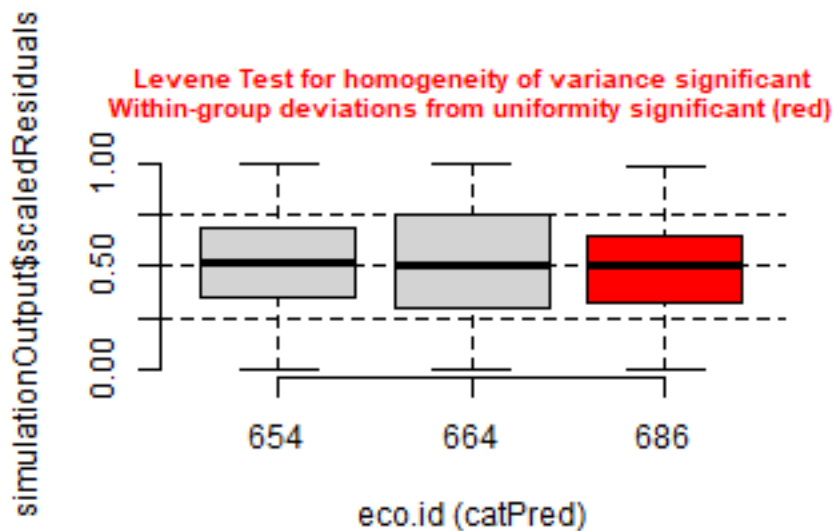
```
plotResiduals(simulation_output_1$scaledResiduals, sites$esy4)
## Warning in ensurePredictor(simulationOutput, predictor):
## DHARMa::ensurePredictor: character string was provided as predictor. DHARMa
## has converted to factor automatically. To remove this warning, please convert
## to factor before attempting to plot with DHARMa.
```



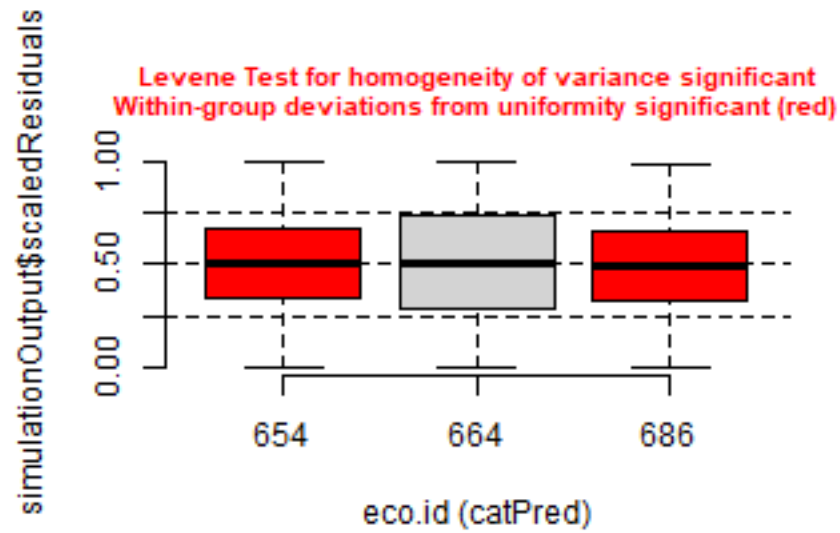

```
plotResiduals(simulation_output_2$scaledResiduals, sites$esy4)
## Warning in ensurePredictor(simulationOutput, predictor):
## DHARMA::ensurePredictor: character string was provided as predictor. DHARMA
## has converted to factor automatically. To remove this warning, please convert
## to factor before attempting to plot with DHARMA.
```



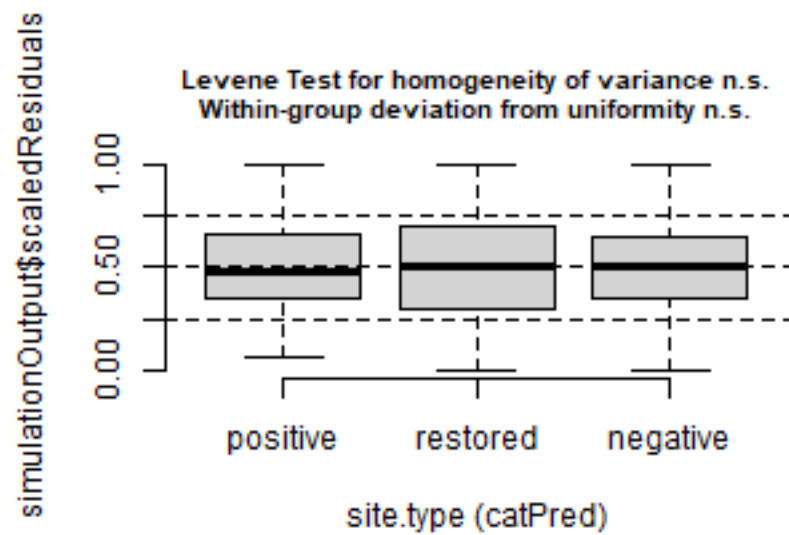
```
plotResiduals(simulation_output_1$scaledResiduals, sites$eco.id)
```



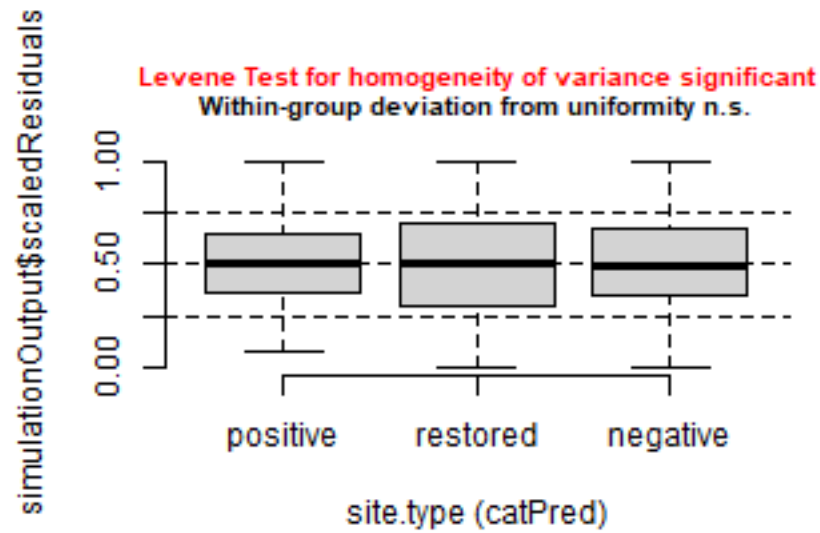
```
plotResiduals(simulation_output_2$scaledResiduals, sites$eco.id)
```



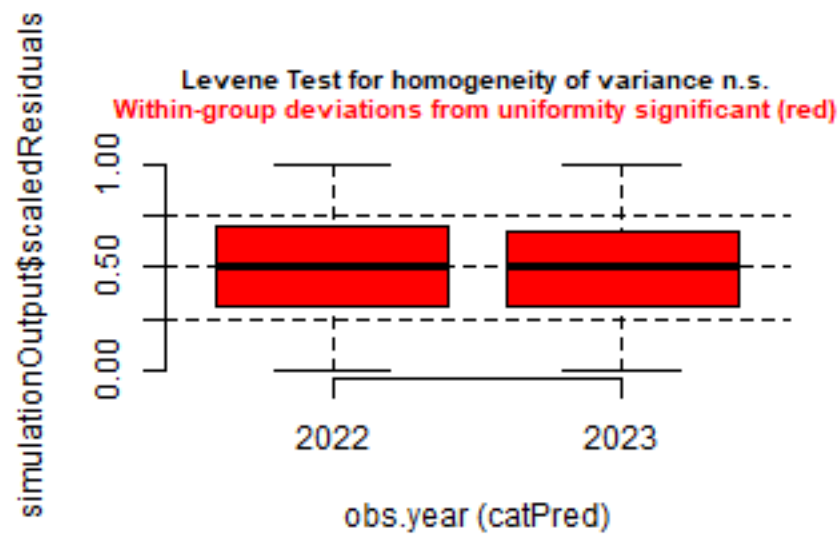
```
plotResiduals(simulation_output_1$scaledResiduals, sites$site.type)
```



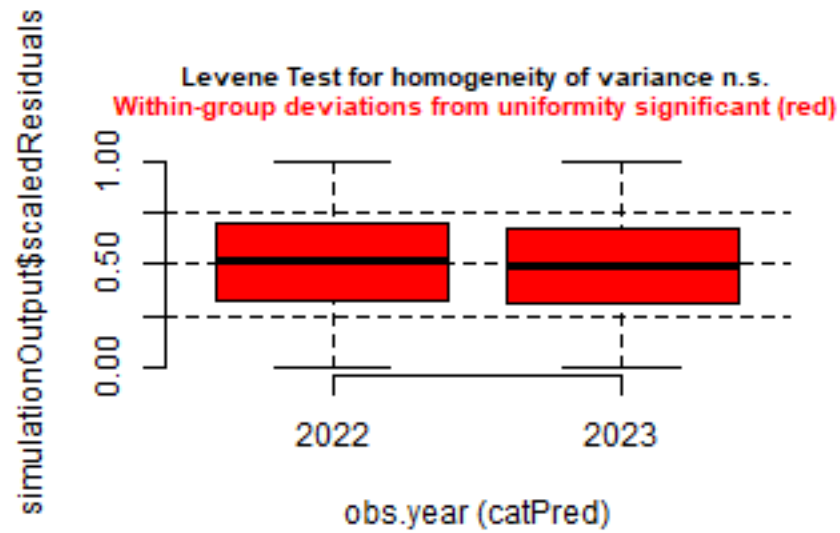
```
plotResiduals(simulation_output_2$scaledResiduals, sites$site.type)
```



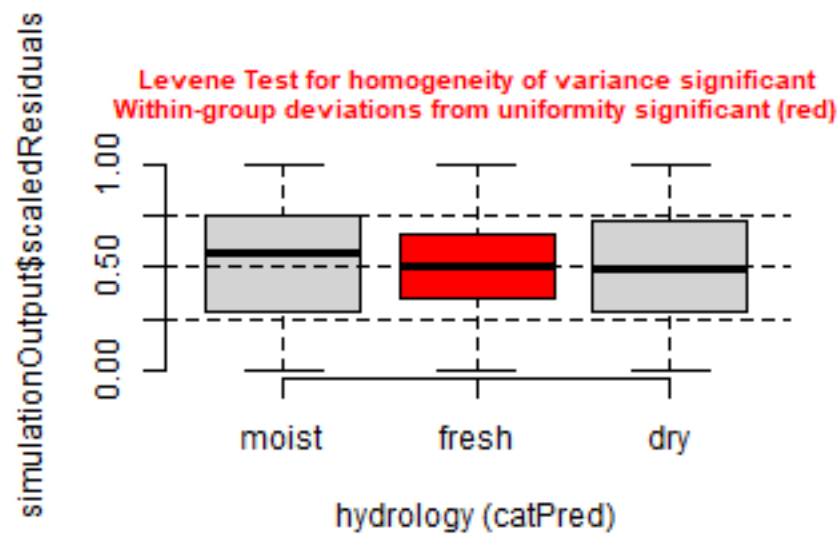
```
plotResiduals(simulation_output_1$scaledResiduals, sites$obs.year)
```



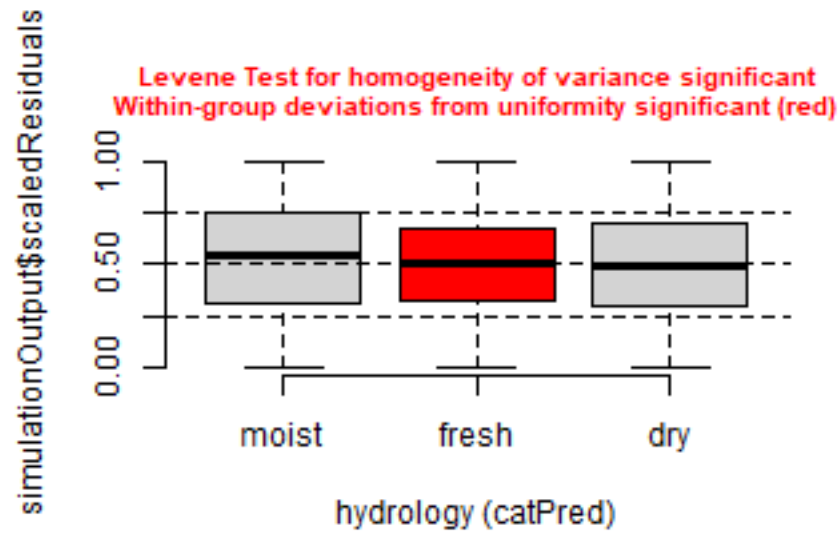
```
plotResiduals(simulation_output_2$scaledResiduals, sites$obs.year)
```



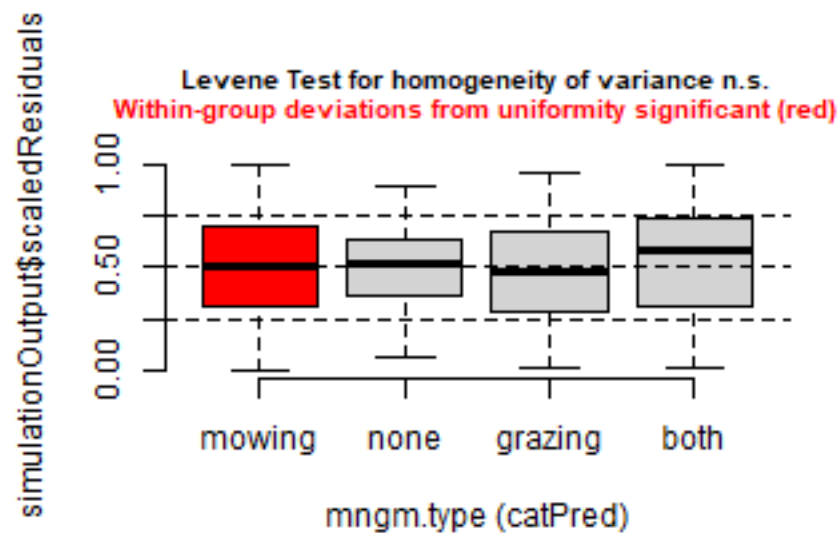
```
plotResiduals(simulation_output_1$scaledResiduals, sites$hydrology)
```



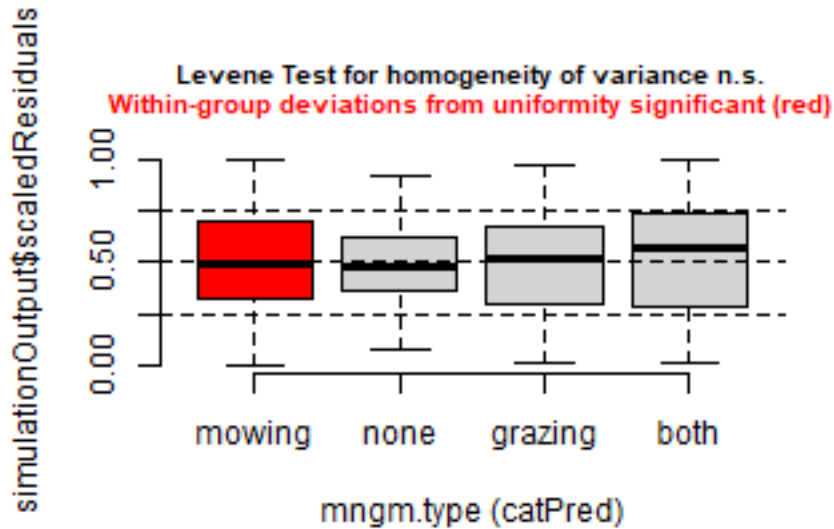
```
plotResiduals(simulation_output_2$scaledResiduals, sites$hydrology)
```



```
plotResiduals(simulation_output_1$scaledResiduals, sites$mngm.type)
```



```
plotResiduals(simulation_output_2$scaledResiduals, sites$mngm.type)
```



Check collinearity part 2 (Step 5)

Remove $VIF > 3$ or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
## Registered S3 method overwritten by 'car':
##   method      from
##   na.action.merMod lme4

##           GVIF Df GVIF^(1/(2*Df))
## hydrology      5.210667 2      1.510857
## site.type      2.217413 2      1.220286
## eco.id         1.038723 2      1.009543
## obs.year       1.059054 1      1.029104
## hydrology:site.type 7.195651 4      1.279775
```

```
car::vif(m_2)
```

```
##           GVIF Df GVIF^(1/(2*Df))
## hydrology     44.913730 2      2.588778
## site.type     30.657624 2      2.353069
## eco.id        9.873142 2      1.772613
## obs.year      4.400523 1      2.097742
## hydrology:site.type 14.709210 4      1.399422
## hydrology:eco.id 15.882181 4      1.412908
## hydrology:obs.year 5.500120 2      1.531415
## site.type:eco.id 30.081477 4      1.530338
## site.type:obs.year 11.383564 2      1.836833
## eco.id:obs.year 9.077332 2      1.735760
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
##           R2m           R2c
```

```
## [1,] 0.3501568 0.717812
MuMIn::r.squaredGLMM(m_2)
##          R2m          R2c
## [1,] 0.4329253 0.7200251
```

AICc

Use AICc and not AIC since ratio $n/K < 40$ Burnham & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMIn::AICc(m_1, m_2) %>%
  arrange(AICc)
##      df      AICc
## m_1 14 1438.043
## m_2 28 5754.476
```

Predicted values

Summary table

```
car::Anova(m_1, type = 3)

## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: sqrt(y)
##              Chisq Df Pr(>Chisq)
## (Intercept)    11803.1769  1 < 2.2e-16 ***
## hydrology       20.9383  2  2.84e-05 ***
## site.type       6.8004  2   0.03337 *
## eco.id          46.0921  2  9.80e-11 ***
## obs.year        0.0504  1   0.82233
## hydrology:site.type  0.9700  4   0.91431
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(m_1)

## Linear mixed model fit by maximum likelihood ['lmerMod']
## Formula: sqrt(y) ~ hydrology * site.type + eco.id + obs.year + (1 | id.site)
## Data: sites
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  1437.4   1499.6   -704.7    1409.4     618
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.8914 -0.4627  0.0220  0.5011  3.1538
##
## Random effects:
## Groups Name Variance Std.Dev.
## id.site (Intercept) 0.4426  0.6653
## Residual          0.3397  0.5829
## Number of obs: 632, groups: id.site, 172
##
## Fixed effects:
##
##              Estimate Std. Error t value
```

```
## (Intercept)          14.803750    0.136261 108.642
## hydrology.L          -0.663776    0.172301  -3.852
## hydrology.Q          -0.451657    0.141559  -3.191
## site.type.L          -0.478638    0.184818  -2.590
## site.type.Q           0.091702    0.121659   0.754
## eco.id664             0.910161    0.139203   6.538
## eco.id686             0.673824    0.138540   4.864
## obs.year2023         -0.025944    0.115539  -0.225
## hydrology.L:site.type.L -0.058405    0.351483  -0.166
## hydrology.Q:site.type.L  0.007854    0.287155   0.027
## hydrology.L:site.type.Q  0.022101    0.232447   0.095
## hydrology.Q:site.type.Q -0.159942    0.190372  -0.840
##
## Correlation of Fixed Effects:
##          (Intr) hydr.L hydr.Q st.t.L st.t.Q ec.664 ec.686 o.2023 h.L:...L
## hydrology.L  0.141
## hydrology.Q  0.120  0.200
## site.type.L -0.073 -0.425 -0.397
## site.type.Q  0.488  0.272  0.217 -0.175
## eco.id664   -0.496  0.096 -0.011 -0.072  0.020
## eco.id686   -0.543  0.052  0.038 -0.045 -0.011  0.502
## obs.yer2023 -0.459 -0.033  0.184 -0.034 -0.039 -0.004  0.031
## hydr1.L:...L -0.290 -0.366 -0.328  0.298 -0.384 -0.016  0.059 -0.014
## hydr1.Q:...L -0.296 -0.340  0.093  0.323 -0.365 -0.067  0.029  0.100  0.239
## hydr1.L:...Q  0.203  0.711  0.194 -0.381  0.198  0.042 -0.021 -0.069 -0.335
## hydr1.Q:...Q  0.060  0.203  0.729 -0.361  0.318 -0.018  0.048  0.172 -0.297
##          h.Q:...L h.L:...Q
## hydrology.L
## hydrology.Q
## site.type.L
## site.type.Q
## eco.id664
## eco.id686
## obs.yer2023
## hydr1.L:...L
## hydr1.Q:...L
## hydr1.L:...Q -0.313
## hydr1.Q:...Q  0.087  0.136
```

Forest plot

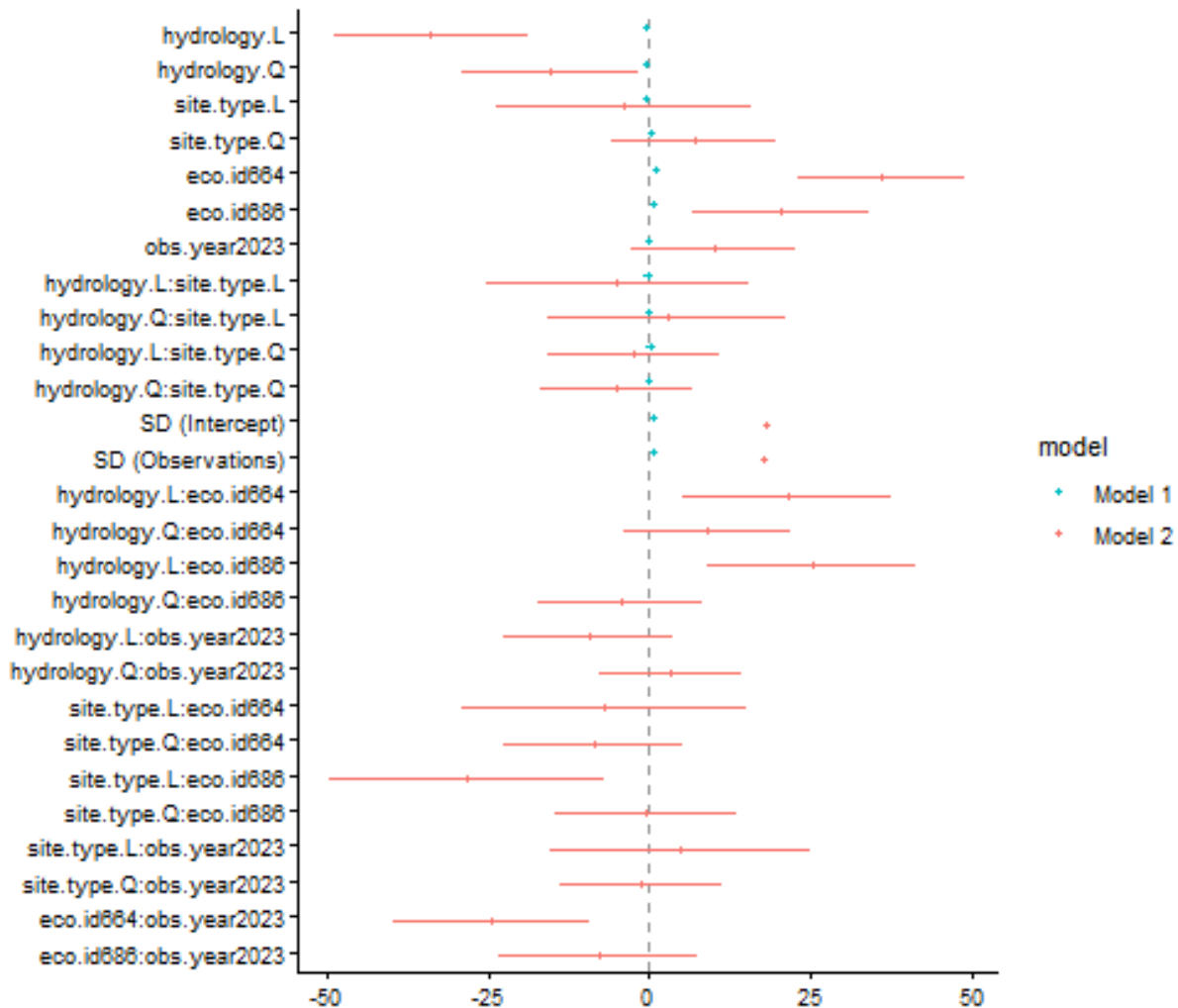
```
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  theme_classic()
```

```
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
```

```
## Warning: Using the `size` aesthetic with geom_segment was deprecated in ggplot2 3.4.0.
## i Please use the `linewidth` aesthetic instead.
```



```
## i The deprecated feature was likely used in the dotwhisker package.
## Please report the issue at <https://github.com/fsolt/dotwhisker/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Effect sizes

Effect sizes of chosen model just to get exact values of means etc. if necessary.

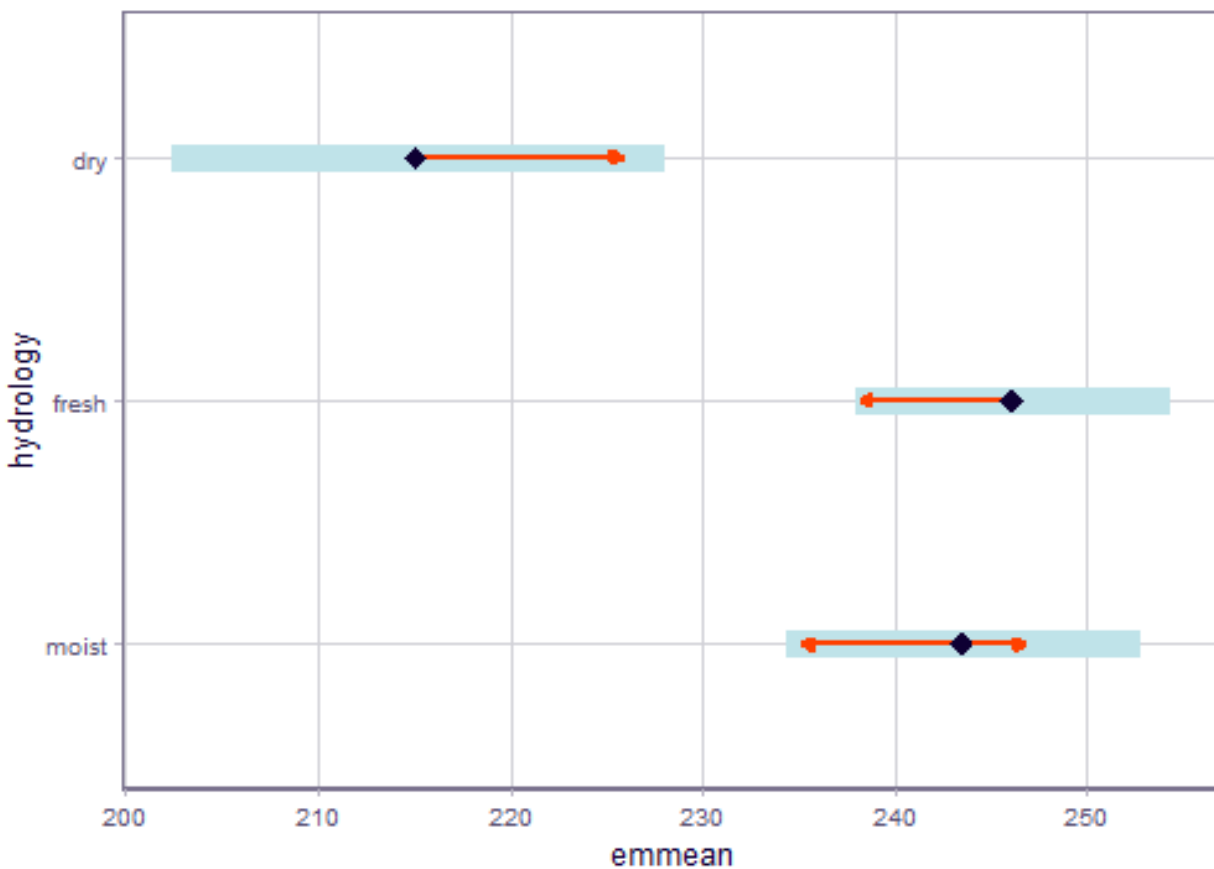
```
(emm <- emmeans(
  m_1,
  revpairwise ~ hydrology,
  type = "response"
))
```

Hydrology

```
## $emmeans
## hydrology response SE df lower.CL upper.CL
```

```
## moist          243 4.66 166      234      253
## fresh          246 3.97  26      238      254
## dry            215 5.95  13      202      228
##
## Results are averaged over the levels of: site.type, eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
## Intervals are back-transformed from the sqrt scale
##
## $contrasts
## contrast      estimate      SE    df t.ratio p.value
## fresh - moist    0.0838 0.197 60.5   0.425 0.9054
## dry - moist     -0.9387 0.252 23.1  -3.722 0.0031
## dry - fresh     -1.0225 0.239 15.7  -4.277 0.0016
##
## Results are averaged over the levels of: site.type, eco.id, obs.year
## Note: contrasts are still on the sqrt scale. Consider using
##       regrid() if you want contrasts of back-transformed estimates.
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
plot(emm, comparison = TRUE)
```



```
(emm <- emmeans(
```

```

m_1,
revpairwise ~ site.type | hydrology,
type = "response"
))

```

Hydrology x site type

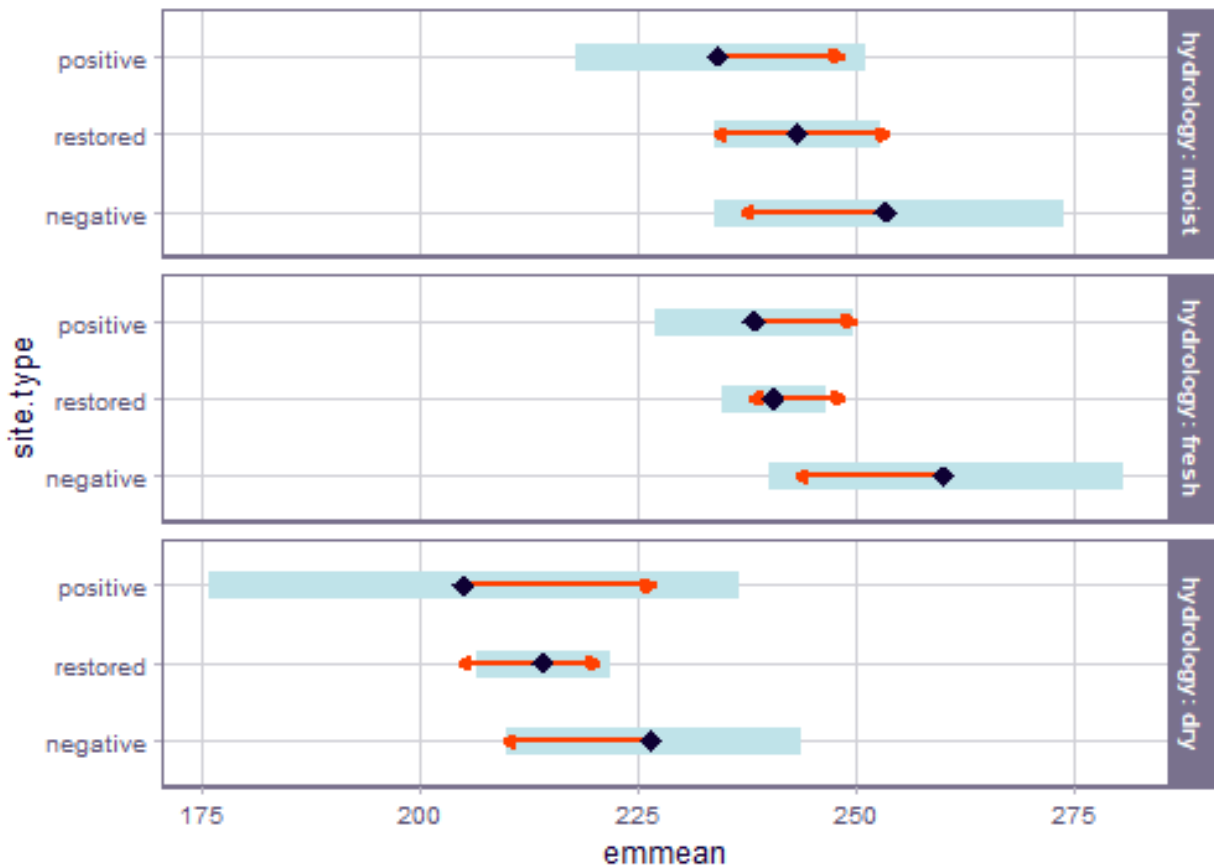
```

## $emmeans
## hydrology = moist:
##   site.type response      SE      df lower.CL upper.CL
##   negative      253 10.20  483.1      234      274
##   restored      243  4.81  177.2      234      253
##   positive      234  8.33   95.6      218      251
##
## hydrology = fresh:
##   site.type response      SE      df lower.CL upper.CL
##   negative      260 10.30 2899.3      240      281
##   restored      240  2.99  177.3      235      246
##   positive      238  5.31   15.1      227      250
##
## hydrology = dry:
##   site.type response      SE      df lower.CL upper.CL
##   negative      226  7.42    8.4      210      244
##   restored      214  3.85  177.2      207      222
##   positive      205 15.50 3926.8      176      236
##
## Results are averaged over the levels of: eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
## Intervals are back-transformed from the sqrt scale
##
## $contrasts
## hydrology = moist:
##   contrast      estimate      SE      df t.ratio p.value
##   restored - negative -0.3202 0.355  356.74  -0.903 0.6390
##   positive - negative -0.6140 0.419  159.77  -1.465 0.3107
##   positive - restored -0.2938 0.313  106.67  -0.937 0.6180
##
## hydrology = fresh:
##   contrast      estimate      SE      df t.ratio p.value
##   restored - negative -0.6152 0.335  974.88  -1.834 0.1590
##   positive - negative -0.6860 0.361   23.98  -1.900 0.1605
##   positive - restored -0.0707 0.197   17.03  -0.358 0.9319
##
## hydrology = dry:
##   contrast      estimate      SE      df t.ratio p.value
##   restored - negative -0.4169 0.279    9.18  -1.493 0.3380
##   positive - negative -0.7308 0.595   11.96  -1.228 0.4600
##   positive - restored -0.3139 0.556 1193.19  -0.564 0.8390
##
## Results are averaged over the levels of: eco.id, obs.year
## Note: contrasts are still on the sqrt scale. Consider using
##       regrid() if you want contrasts of back-transformed estimates.
## Degrees-of-freedom method: kenward-roger

```

```
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
plot(emm, comparison = TRUE)
```



Session info

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
## LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8 LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats graphics grDevices utils datasets methods base
##
## other attached packages:
```

```

## [1] emmeans_2.0.3    DHARMA_0.5.0    patchwork_1.3.2  ggbeeswarm_0.7.3
## [5] lubridate_1.9.5   forcats_1.0.1   stringr_1.6.0    dplyr_1.2.1
## [9] purrr_1.2.2       readr_2.2.0     tidyr_1.3.2      tibble_3.3.1
## [13] ggplot2_4.0.3     tidyverse_2.0.0 here_1.0.2
##
## loaded via a namespace (and not attached):
## [1] Rdpack_2.6.6      gridExtra_2.3     sandwich_3.1-1
## [4] rlang_1.2.0       magrittr_2.0.5    multcomp_1.4-30
## [7] otel_0.2.0        compiler_4.5.2    mgcv_1.9-3
## [10] vctrs_0.7.3       pkgconfig_2.0.3   crayon_1.5.3
## [13] fastmap_1.2.0     backports_1.5.1   labeling_0.4.3
## [16] utf8_1.2.6        ggstance_0.3.7    promises_1.5.0
## [19] rmarkdown_2.31    tzdb_0.5.0        nloptr_2.2.1
## [22] bit_4.6.0         xfun_0.58         later_1.4.8
## [25] broom_1.0.13      parallel_4.5.2    R6_2.6.1
## [28] gap.datasets_0.0.6 stringi_1.8.7      qgam_2.0.0
## [31] RColorBrewer_1.1-3 car_3.1-5          boot_1.3-32
## [34] estimability_1.5.1 Rcpp_1.1.1-1.1     iterators_1.0.14
## [37] knitr_1.51        zoo_1.8-15        parameters_0.29.1
## [40] httpuv_1.6.17     Matrix_1.7-4      splines_4.5.2
## [43] timechange_0.4.0  tidysselect_1.2.1 rstudioapi_0.18.0
## [46] abind_1.4-8       yaml_2.3.12       MuMIn_1.48.19
## [49] doParallel_1.0.17 codetools_0.2-20   lattice_0.22-7
## [52] plyr_1.8.9        bayestestR_0.18.1 shiny_1.13.0
## [55] withr_3.0.2       S7_0.2.2          evaluate_1.0.5
## [58] marginaleffects_0.32.0 survival_3.8-3     pillar_1.11.1
## [61] gap_1.15.2        carData_3.0-6     foreach_1.5.2
## [64] stats4_4.5.2      insight_1.5.1     reformulas_0.4.4
## [67] generics_0.1.4    vroom_1.7.1       rprojroot_2.1.1
## [70] hms_1.1.4         scales_1.4.0      minqa_1.2.8
## [73] xtable_1.8-8      glue_1.8.1        tools_4.5.2
## [76] data.table_1.18.4 lme4_2.0-1        mvtnorm_1.4-0
## [79] grid_4.5.2        rbibutils_2.4.1   datawizard_1.3.1
## [82] nlme_3.1-168      Rmisc_1.5.1       performance_0.17.0
## [85] beeswarm_0.4.0    vipor_0.4.7       Formula_1.2-5
## [88] cli_3.6.6         gtable_0.3.6      digest_0.6.39
## [91] pbkrtest_0.5.5    TH.data_1.1-5     farver_2.1.2
## [94] htmltools_0.5.9   lifecycle_1.0.5   mime_0.13
## [97] bit64_4.8.2       dotwhisker_0.8.4  MASS_7.3-65

```